Overnight returns, daytime reversals, and anchoring bias

Donghoon Kim[1], Jihoon Goh[2]

[1] College of Business, Korea Advanced Institute of Science and Technology, 85 Hoegiro, Dongdaemoon-gu, Seoul, 02455, Republic of Korea; Tel: +82 2 958 3552; E-mail: kdh9406@kaist.ac.kr
[2] (Corresponding author) Pusan National University, 2, Busandaehak-ro 63beon-gil, Geumjeong-gu, Busan, 46241, Republic of Korea, Tel:+82-51-510-2566, jihoongoh@pusan.ac.kr

* This work was supported by a New Faculty Research Grant of Pusan National University, 2023.

Preprint not peer reviewed

This preprint research paper has not been peer reviewed. Electronic copy available at: https://ssrn.com/abstract=4482812

Overnight returns, daytime reversals, and anchoring bias

June 2023

Abstract

This paper examines the dynamics between overnight noise traders and daytime arbitrageurs, measured by AB_NR from Akbas et al. (2022), which significantly differ in the predictability of future returns among the nearness to their 52-week high prices. If stocks are far below their 52-week highs, there is a perception of greater potential for price increases, leading to tremendous upward pressure on prices by overnight noise traders. As a result, daytime arbitrageurs overcorrect, leading to high AB_NR stocks offering the potential for increased future returns. Our findings provide a fresh perspective on the psychological barrier of investors to intense tug-of-war.

JEL classification: G11; G12; G14

Keywords: Overnight return; Daytime reversal; Arbitrageurs; 52-week high; Anchoring bias

Preprint not peer reviewed

This preprint research paper has not been peer reviewed. Electronic copy available at: https://ssrn.com/abstract=4482812

1. Introduction

The dynamics of financial markets have been an important subject of study for researchers and practitioners alike. Especially the release of major economic news and information outside regular stock exchange trading hours has become increasingly common. The different behaviors among overnight and daytime traders plays a vital role in comprehending the dynamics of financial markets.

Recently, the daily tug-of-war concept has received considerable attention from researchers. Lou et al. (2019) find that two distinct clientele, individual investors and institutional investors, create a tug-of-war, resulting in return reversals across overnight and daytime periods. Akbas et al. (2022) document that stocks with intense daily tug-of-war are underpriced because daytime arbitrageurs overcorrect positive overnight returns. To gauge the intensity of the tug-of-war, they quantify it by calculating the abnormal frequency of the occurrence of positive overnight returns that are succeeded by negative reversals during regular trading hours, which is referred to as AB_NR. The study indicates that daytime arbitragers tend to disregard the portion of overnight returns that may be associated with fundamental changes. They demonstrate that daytime arbitrageurs persistently exhibit overcorrection behavior in response to consecutive positive overnight returns

This study examines the predictive power of daily tug-of-war measured by AB_NR is primarily strong among stocks far from their 52-week high prices. As suggested by Kahneman and Tversky (1974), people use a reference point, known as an "anchor," and make adjustments based on it. According to George and Hwang (2004), the 52-week high, which represents the highest price at which a stock has traded in the past year, serves as a reference point for investors to evaluate the potential price movement of a stock. Birru (2015) finds that investors view the 52-week high as an upper limit, expecting decline when the stock is near it and an increase when it is far. Instead of estimating the stock's value, investors evaluate whether new information is already incorporated or if there is "room to run" for further price changes. When a stock is close to its 52-week high, investors may believe that positive news is already incorporated in the price, limiting aggressive bidding and dampening price reactions to positive events.

The anchoring bias on the 52-week high is a valuable consideration when exploring the phenomenon of

Preprint not peer reviewed

This preprint research paper has not been peer reviewed. Electronic copy available at: https://ssrn.com/abstract=4482812

intense tug-of-war. Nearness to the 52-week high price might influence the overnight noise traders who regard the 52-week high price as an anchor point. We hypothesize the existence of intense tug-of-war dynamics between noise traders during overnight period and arbitrageurs during daytime period are largely affected by the nearness to the 52-week high price. In situations where stocks are trading significantly below their 52-week highs, overnight noise traders apply substantial upward pressure on these stocks due to their perception of greater potential for price increases. Meanwhile, daytime arbitrageurs tend to overcorrect for the overpricing of these stocks, resulting in a relative underpricing of high AB_NR stocks compared to low AB_NR stocks. As a result of this overcorrection, high AB_NR stocks become undervalued and offer the potential for higher future returns. Conversely, when stocks are trading near their 52-week highs, overnight noise traders perceive less "room to go" and exert less pressure on prices. This results in daytime arbitrageurs overcorrecting less and high AB_NR stocks being less underpriced, which may lead to weaker return predictability.

To test our hypothesis, we employ a 52-week high price from George and Hwang (2004) as the reference price for the anchoring bias and AB_NR from Akbas et al. (2022) for the intense tug-of-war. The stocks are divided into quintiles based on their proximity to the 52-week high (NH), and they are subsequently sorted independently according to their AB_NR levels. The results indicate that high AB_NR stocks underperform, particularly those far from their 52-week prices. The AB_NR premium is significant for stocks far from the 52-week high (1.08% with a t-statistic of 5.50) but only marginally significant for stocks near it (0.33% with a t-statistic 2.83). We also conduct cross-sectional regressions based on the Fama and MacBeth (1973) methodology, controlling for other predictors of stock returns. The analysis shows significant negative coefficients on the interaction between proximity to the 52-week high and AB_NR (-0.113 with t-statistic -4.88), indicating that intense tug-of-war strategies with stocks far from their 52-week high outperform those near it. Our findings are further supported during high sentiment periods, where optimistic noise traders have a greater impact on stocks far from their 52-week highs than low sentiment periods.

This study reinforces the assumptions made in previous research, such as Lou et al. (2019) or Akbas et al. (2022), regarding the reactions of investors to overnight returns. By exploring the connection between overnight and daytime returns in the context of investors' anchoring bias, our study offers a novel perspective on the

Preprint not peer reviewed

This preprint research paper has not been peer reviewed. Electronic copy available at: https://ssrn.com/abstract=4482812

psychological hurdle faced by noise investors during overnight periods.

2. Data and Variables

Our analysis is based on a sample of publicly traded stocks with share codes 10 and 11 listed on NYSE, AMEX, and NASDAQ from May 1993 to December 2020. The daily and monthly return data for each stock is obtained from the Center for Research in Security Prices (CRSP) database.

We follow Akbas et al. (2022) in calculating the daily intensity of tug-of-war using $NR_{i,t}$, which captures the occurrence rate of negative daytime reversals. We normalize $NR_{i,t}$ by the average over the preceding 12 months to obtain AB_NR ($AB_NR_{i,t}$), representing abnormal intensity of tug-of-war. In line with Akbas et al. (2022), our focus is on negative reversals due to their demonstrated return predictability, while daily positive reversals do not exhibit the same predictability. In order to gain a comprehensive understanding of the impact of daily tug-of-war intensity, we also examine the predictive capability of both the level and abnormal levels, specifically with regard to positive reversals ($PR_{i,t}$).

The anchoring variable used in our study is consistent with the approach George and Hwang (2004) adopted, which is based on the ratio of a stock's current price to its 52-week high price. The 52-week high price is calculated using the highest daily closing prices over the previous 52 weeks, adjusted for any distributions such as stock dividends or splits, using the CRSP price adjustment factor. We calculate the nearness to the 52-week high price (NH) by dividing the stock price by the 52-week high price.

As part of our analysis, we include several control variables to account for potential confounding factors that may impact our results. Specifically, we consider the beta (BETA), size (ME), book-to-market ratio (BM), momentum (MOM), idiosyncratic volatility (IVOL), turnover (TURN), and illiquidity (ILLIQ). These variables are commonly used in empirical finance literature as measures of the cross-section of stock return predictors. By controlling for these factors, we aim to isolate the effect of anchoring bias on daily tug-of-war intensity on stock returns. Table 1 presents the summary statistics of our study. Panel A reports the time-series average of each variable sorted by the AB_NR deciles, and Panel B represents the correlation of the variables.

Preprint not peer reviewed

This preprint research paper has not been peer reviewed. Electronic copy available at: https://ssrn.com/abstract=4482812

[Table 1 about here]

3. Empirical Results

3.1. Intense tug-of-war and the nearness to the 52-week high

We first analyze to determine if the AB_NR premium from Akbas et al. (2022) is present in our sample of stocks by ranking individual stocks according to their AB_NR values and creating decile portfolios based on those rankings. The decile portfolios, whether equally or value weighted, are held for a one-month period. The AB_NR premium is determined by calculating the return difference between the lowest and highest deciles. Our results, presented in Table 2, indicate that the returns of the AB_NR strategy are significant for equally weighted but insignificant for the value-weighted, with returns of 0.83% and 0.27%, respectively.[1]

[Table 2 about here]

To conduct a formal analysis of our hypotheses, we employ a double-sorting procedure to create AB_NR premiums specifically for sorted quintile portfolios based on their 52-week high prices. Initially, we divide the entire sample of stocks into quintile portfolios based on their NH. Subsequently, we categorize individual stocks into quintiles based on their AB_NR values, and compute equal-weighted returns for each of the resulting 25 portfolios. We then calculate the AB_NR premia for the NH quintiles by comparing the returns of the lowest and highest AB_NR quintiles. Table 3 reports the average returns and return spread of AB_NR for quintiles far below their 52-week high to near their 52-week high executives.

[Table 3 about here]

The far below their 52-week high group exhibits average returns of 0.07% for the lowest AB_NR quintile and 1.14% for the highest AB_NR quintile, for the equal-weighted portfolios. They are 0.82% and 1.15% for the near their 52-week high group, corresponding AB_NR premia of 1.08% (t-statistic = 5.50) and 0.33% (t-statistic = 2.83), respectively. The difference in AB_NR premia between the lowest and highest NH quintiles is -0.75% with

[1] This is consistent with the results from Akbas et al. (2022).

Preprint not peer reviewed

This preprint research paper has not been peer reviewed. Electronic copy available at: https://ssrn.com/abstract=4482812

a t-statistic of -4.23, which is enormously significant. Furthermore, when adjusting the returns of the far below their 52-week high group by the Carhart (1997) four-factor model and the Fama and French (2015) five-factor model, the risk-adjusted AB_NR premia consistently exhibit positive values with high statistical significance. In contrast, the AB_NR premia of the near 52-week high group are always weakly significant under risk adjustments. Thus, the evidence that the intense tug-of-war is primarily concentrated among stocks far below their 52-week high aligns with our hypothesis.

3.2. Fama and Macbeth (1973) cross-sectional regression

To further test our hypotheses, we perform the Fama and MacBeth (1973) cross-sectional regressions to control for other factors that may influence stock returns. Specifically, for each month, we conduct a cross-sectional regression as follows:

$$\mathrm{RET}_{i,t+1} = \lambda_{0,t} + \lambda_{1,t} \cdot \mathrm{AB_NR}_{i,t} + \lambda_{2,t} \cdot \mathrm{NH}_{i,t} + \lambda_{3,t} \cdot \mathrm{AB_NR}_{i,t} \times \mathrm{NH}_{i,t} + \psi_t \cdot \mathrm{Z}_{i,t} \qquad (1)$$

where the dependent variable is the stock return in month t+1 ($\mathrm{RET}_{i,t+1}$), and the independent variables include the stock's AB_NR ($\mathrm{AB_NR}_{i,t}$), nearness to the 52-week high ($\mathrm{NH}_{i,t}$), and the interaction term between AB_NR and the nearness to their 52-week high variable ($\mathrm{AB_NR}_{i,t} \times \mathrm{NH}_{i,t}$). Additionally, $\mathrm{Z}_{i,t}$ is the vector of several other control variables, including market beta, past returns, momentum, market capitalization, book-to-market ratio, illiquidity, turnover, and idiosyncratic volatility. We perform the regression for every month and estimate the time-series average of coefficients λ and ψ with the corresponding t-statistics using the method of robust Newey and West (1987) standard errors with lag 12. Based on our hypothesis, we expect a negative and significant coefficient for the interaction term, indicating that the intense tug-of-war strategy among stocks near their 52-week high will underperform the intense tug-of-war strategy among stocks far below their 52-week high.

[Table 4 about here]

Table 4 represents the results of cross-sectional regressions of various models. When we run the regression with control variables (Model 6), we find that the average coefficient for the interaction term is -0.113 with a t-statistic of -4.88, supporting our hypothesis. We can interpret the coefficients as follows. The coefficient of AB_NR,

Preprint not peer reviewed

This preprint research paper has not been peer reviewed. Electronic copy available at: https://ssrn.com/abstract=4482812

which represents the return predictability of AB_NR, for stocks near 52-week highs (1 sigma above the mean) is 0.029 (0.142 – 0.113), which is insignificant. On the other hand, the coefficient of AB_NR for stocks far from 52-week highs (1 sigma below the mean) is 0.255 (0.142 + 0.113), which is strongly significant. The average coefficient remains consistently negative and significant for other regression specifications (Models 3 to 5).

3.3. Time-varying of intense tug-of-war and the nearness to the 52-week high

In this section, we examine the predictive power of intense tug-of-war and the proximity of stock prices to their 52-week highs concerning market sentiment. Previous research highlights the impact of sentiment on investor behavior and market mispricing using the sentiment index proposed by Baker and Wurgler (2006). Baker and Wurgler (2006) argue that stocks that are hard to value and exhibit limited arbitrage opportunities are more vulnerable to shifts in market sentiment. Research also indicates that mispricing tends to be more prevalent during periods of high market sentiment and sentiment has been found to weaken the relationship between the mean and variance of the market portfolio. (Aboody et al., 2018; Baker and Wurgler, 2006; Yu and Yuan., 2011; Stambaugh et al., 2012). Building on these findings, Akbas et al. (2022) suggest that daytime arbitrage traders tend to attribute prolonged sequences of positive overnight returns of noise traders during high sentiment periods. This belief in noise trading as the cause of overnight price pressure leads to even greater overcorrection, resulting in a stronger predictive relationship for AB_NR. Combining the literature with our hypothesis, we expect that stocks trading significantly far below their 52-week highs are perceived to have potential price increases, leading to upward pressure by overnight noise traders during high sentiment periods and subsequent overcorrection by daytime arbitrageurs. This results in the undervaluation of high AB_NR stocks compared to low sentiment periods, offering more potential for future returns.

To explore the relationship between sentiment and the intense tug-of-war with investors' anchoring bias, we re-examine the bivariate sort portfolio analysis in Table 3 during high and low sentiment states, using the Baker and Wurgler (BW) sentiment index.[2] Following the approach of Baker and Wurgler (2007) and Stambaugh et al.

[2] The BW index is derived from various proxies and adjusted for business cycle variables by following Baker and Wurgler (2006).

Preprint not peer reviewed

This preprint research paper has not been peer reviewed. Electronic copy available at: https://ssrn.com/abstract=4482812

(2012), we define a high (low) sentiment month when the BW index exceeds (falls below) the median value, denoting these states as 1 (0).

[Table 5 about here]

Table 5 reveals two notable findings. First, we observe more significant differences in the long-short portfolio of AB_NR quintiles across all NH quintiles. This aligns with our narrative that during high sentiment periods, overnight noise traders become more optimistic, leading to a greater occurrence of daytime overcorrection. These results are also consistent with the findings of Akbas et al. (2022). Secondly, we find that there is a significant difference in AB_NR premia between stocks near and far from their 52-week price during high sentiment (-0.94% with t-statistic -3.14), while this significance diminishes during low sentiment periods (-0.48% with t-statistic -1.72). Our findings provide additional support by demonstrating that the impact of optimistic noise traders, leading to positive price pressure, is more significant when market sentiment is high compared to low sentiment periods. This consistency highlights the influence of high sentiment in amplifying the effect on stocks that are far from their 52-week highs.

4. Conclusion

Our paper sheds light on the role of anchoring bias in the tug-of-war between overnight and daytime traders. Our findings reveal that the predictive power of the intense daily tug-of-war, as measured by AB_NR, is more pronounced among stocks that are trading far below their 52-week highs. This suggests that noise traders may be influenced by anchoring bias, and they may generate more (less) upward price pressure for the positive information of stocks followed by overcorrection of arbitrageurs when their prices are far from (near) their 52-week highs. Our findings contribute a fresh perspective on the psychological barrier that investors face when dealing with overnight returns. Moreover, our study emphasizes the significance of accounting for anchoring bias in comprehending the return dynamics of stock markets.

Preprint not peer reviewed

This preprint research paper has not been peer reviewed. Electronic copy available at: https://ssrn.com/abstract=4482812

Appendix. Variable Descriptions

$RET_CO_{i,d}$ represents the overnight return from the previous day's closing price to the opening price on the current day and $RET_OC_{i,d}$ represents the daytime return from the opening price to the closing price on the current day, by following Lou et al. (2019).

$$RET_OC_{i,d} = \frac{prc_{i,d}^{close}}{prc_{i,d}^{open}} - 1 \text{ ,} RET_CO_{i,d} = \frac{(1 + RET_{i,d})}{(1 + RET_OC_{i,d})} - 1 \qquad (A1)$$

where $RET_{i,d}$ is the standard daily close-to-close return, $prc_{i,d}^{close}$ is the previous day's closing price and $prc_{i,d}^{open}$ is the current day's opening price.

$NR_{i,t}$ is a variable that measures the proportion of days in a given month for firm i in which there is a negative daytime reversal, by following Akbas et al. (2022). It is computed by taking the number of days with a positive overnight return ($RET_CO_{i,d} > 0$) and a negative daytime return ($RET_OC_{i,d} < 0$) and dividing it by the total number of trading days for the firm in that month. Similarly, $PR_{i,t}$ is a variable that measures the proportion of days in a given month for firm i in which there is a positive daytime reversal, by taking the number of days with a negative overnight return ($RET_CO_{i,d} <0$) and a positive daytime return ($RET_OC_{i,d} > 0$) and dividing it by the total number of trading days for the firm in that month To calculate $NR_{i,t}$ ($PR_{i,t}$), we need to have at least 15 trading days in that month for the firm. $AB_NR_{i,t}$ ($AB_NR_{i,t}$) is the standardized value of $NR_{i,t}$ ($PR_{i,t}$) to the average $NR_{i,t}$ ($PR_{i,t}$) over the past 12 months.

$RET_{i,t}$ is the contemporaneous monthly return from CRSP in month t. This also refers to using the return of a stock in the previous month to predict its future performance by following Jegadeesh (1990).

$RETCO_M_{i,t}$ is the compounded daily overnight return during month t and $RETOC_M_{i,t}$ is the compounded daily daytime return during month t.

$NH_{i,t}$ represents the ratio of the stock price of firm i at the end of month t ($prc_{i,t}$) to its 52-week high price ($52high_{i,t}$) by following George and Hwang (2004). The 52-week high price is defined as the highest daily closing price of firm i from the beginning of month t-11 to the end of month t.

$$NH_{i,t} = \frac{prc_{i,t}}{52high_{i,t}} \qquad (A2)$$

$BM_{i,t}$ is the natural logged value of the firm's book-to-market ratio from the fiscal year ending at least six months prior to month t and $ME_{i,t}$ is the natural logged value of the firm's market capitalization is defined as the total number of shares outstanding for firm i multiplied by the share price, on the last day of month t by following Fama and French (1992).

$BETA_{i,t}$ is stock i's market beta in month t, which is computed by regressing stock i's daily returns on the current daily

Preprint not peer reviewed

This preprint research paper has not been peer reviewed. Electronic copy available at: https://ssrn.com/abstract=4482812

market return, as well as the lag and lead market returns to control for nonsynchronous trading, following the methods of Scholes and Williams (1977) and Dimson (1979).

$$\mathrm{RET}_{i,d} - \mathrm{RETt}_{f,d} = \alpha_i + \beta_{1,i} \cdot \mathrm{mktrf}_{d-1} + \beta_{2,i} \cdot \mathrm{mktrf}_d + \beta_{3,i} \cdot \mathrm{mktrf}_{d+1} + \epsilon_{i,d} \qquad \text{(A3)}$$

where $\mathrm{RET}_{i,d}$ is the return on stock i on day d and $\mathrm{ret}_{f,d}$ is the T-Bill return on day d. mktrf_{d-1}, mktrf_d, and mktrf_{d+1} are the excess market return from the previous, current, and future relative to risk-free rate respectively. BETA of stock i at the end of the month t is calculated by $\widehat{\beta_{1,i}} + \widehat{\beta_{2,i}} + \widehat{\beta_{3,i}}$.

$\mathrm{IVOL}_{i,t}$ is calculated as the standard deviation of the daily residuals obtained from the Fama and French (1993) three-factor model for that month, by following Ang et al. (2006).

$$\mathrm{RET}_{i,d} - \mathrm{RETt}_{f,d} = \alpha_i + \beta_{1,i} \cdot \mathrm{mktrf}_d + \beta_{2,i} \cdot \mathrm{SMB}_d + \beta_{1,i} \cdot \mathrm{HML}_d + \epsilon_{i,d} \qquad \text{(A4)}$$

where $\mathrm{RET}_{i,d}$ is the return on stock i on day d and $\mathrm{RET}_{f,d}$ is the T-Bill return on day d. mktrf_d, SMB_d, and HML_d are daily three-factors from Fama and French (1993). $\mathrm{IVOL}_{i,t}$ of stock i at the end of the month t is defined as the standard deviation of daily residuals, $\epsilon_{i,d}$ in month t.

$\mathrm{ILLIQ}_{i,t}$ is the firm's illiquidity measure by following Amihud (2002).

$$\mathrm{ILLIQ}_{i,t} = \frac{1}{D_{i,t}} \sum_{d=1}^{D_{i,t}} \frac{|R_{i,d}|}{\mathrm{dollarvolume}_{i,d,}} \qquad \text{(A5)}$$

where $R_{i,d}$ is the stock return for firm i on day d of month t, $\mathrm{dollarvolume}_{i,d}$ is the corresponding daily volume in dollars, and $D_{i,t}$ is the number of days in month t for which data are available.

$\mathrm{MOM}_{i,t}$ is the intermediate-term of momentum, which is calculated by the past 11-month returns of each stock, but with a one-month lag, by following Jegadeesh and Titman (1993).

$\mathrm{TURN}_{i,t}$ is the monthly share turnover of a company is determined by dividing the trading volume (number of shares traded) in a given month by the total number of outstanding shares.

Preprint not peer reviewed

This preprint research paper has not been peer reviewed. Electronic copy available at: https://ssrn.com/abstract=4482812

References

Aboody, D., Even-Tov, O., Lehavy, R., Trueman, B., 2018. Overnight returns and firm-specific investor sentiment. Journal of Financial and Quantitative Analysis 53, 485-505

Akbas, F., Boehmer, E., Jiang, C., Koch, P.D., 2022. Overnight returns, daytime reversals, and future stock returns. Journal of Financial Economics 145, 850-875

Amihud, Y., 2002. Illiquidity and stock returns: cross-section and time-series effects. Journal of financial markets 5, 31-56

Ang, A., Hodrick, R.J., Xing, Y., Zhang, X., 2006. The cross-section of volatility and expected returns. The journal of finance 61, 259-299

Baker, M., Wurgler, J., 2006. Investor sentiment and the cross-section of stock returns. The journal of Finance 61, 1645-1680

Baker, M., Wurgler, J., 2007. Investor sentiment in the stock market. Journal of economic perspectives 21, 129-151

Birru, J., 2015. Confusion of confusions: A test of the disposition effect and momentum. The Review of Financial Studies 28, 1849-1873

Carhart, M.M., 1997. On persistence in mutual fund performance. The Journal of finance 52, 57-82

Dimson, E., 1979. Risk measurement when shares are subject to infrequent trading. Journal of Financial Economics 7, 197-226

Fama, E.F., French, K.R., 1992. The cross-section of expected stock returns. the Journal of Finance 47, 427-465

Fama, E.F., French, K.R., 1993. Common risk factors in the returns on stocks and bonds. Journal of financial economics 33, 3-56

Fama, E.F., French, K.R., 2015. A five-factor asset pricing model. Journal of financial economics 116, 1-22

Fama, E.F., MacBeth, J.D., 1973. Risk, return, and equilibrium: Empirical tests. Journal of political economy 81, 607-636

George, T.J., Hwang, C.Y., 2004. The 52-week high and momentum investing. The Journal of Finance 59, 2145-2176

Jegadeesh, N., 1990. Evidence of predictable behavior of security returns. The Journal of finance 45, 881-898

Jegadeesh, N., Titman, S., 1993. Returns to buying winners and selling losers: Implications for stock market efficiency. The Journal of finance 48, 65-91

Lou, D., Polk, C., Skouras, S., 2019. A tug of war: Overnight versus intraday expected returns. Journal of Financial Economics 134, 192-213

Newey, W.K., West, K.D., 1987. Hypothesis testing with efficient method of moments estimation. International Economic Review, 777-787

Scholes, M., Williams, J., 1977. Estimating betas from nonsynchronous data. Journal of financial economics 5, 309-327

Stambaugh, R.F., Yu, J., Yuan, Y., 2012. The short of it: Investor sentiment and anomalies. Journal of Financial Economics 104, 288-302

Tversky, A., Kahneman, D., 1974. Judgment under Uncertainty: Heuristics and Biases: Biases in judgments reveal some heuristics of thinking under uncertainty. science 185, 1124-1131

Yu, J., Yuan, Y., 2011. Investor sentiment and the mean–variance relation. Journal of Financial Economics 100, 367-381

Preprint not peer reviewed

This preprint research paper has not been peer reviewed. Electronic copy available at: https://ssrn.com/abstract=4482812

Table 1. Summary Statistics

This table presents the summary statistics of our study. Panel A reports the time-series average of each variable for the decile portfolios sorted by $AB_NR_{i,t}$ from Akbas et al. (2022). Panel B represents the time-series average of cross-sectional correlations of the variables. $AB_NR_{i,t}$ ($AB_PR_{i,t}$) is the rate of negative (positive) daytime reversal during the trading day, scaled by month's trading day and normalized from past 12-month average. $RET_{i,t}$ is stock i's return in month t. $RETCO_M_{i,t}$ ($RETOC_M_{i,t}$) is the compounded daily overnight (daytime). $NH_{i,t}$ represents the ratio of the stock price of firm i at the end of month t ($prc_{i,t}$) to its 52-week high price ($52high_{i,t}$). $BM_{i,t}$ is the natural logarithm of stock i's book value of equity at the end of the fiscal year ending in calendar year y-1, divided by its market value of equity at the end of December in year y-1. $MOM_{i,t}$ is stock i's cumulative return from month t-12 to month t-2. $ME_{i,t}$ is the natural logarithm of stock i's market capitalization in month t. $ILLIQ_{i,t}$ is the time-series average of stock i's absolute daily return divided by daily dollar trading volume over month t. Sample period is from May 1993 to December 2020.

Panel A: Summary Statistics

AB_NR decile	AB_NR	AB_PR	RET	RETOC_M	RETCO_M	NH	ME	BM	MOM	ILLIQ
1 (Low)	0.27	1.23	0.04	0.13	-0.07	0.78	324.95	0.84	0.08	9.86
2	0.56	1.18	0.04	0.09	-0.04	0.77	406.14	0.71	0.10	2.68
3	0.71	1.12	0.03	0.06	-0.02	0.76	426.82	0.69	0.11	2.19
4	0.83	1.07	0.03	0.04	-0.01	0.76	448.08	0.68	0.13	2.13
5	0.93	1.03	0.02	0.02	0.00	0.76	448.53	0.68	0.14	1.72
6	1.03	0.99	0.01	0.00	0.02	0.76	432.77	0.67	0.15	1.77
7	1.14	0.95	0.01	-0.02	0.03	0.76	435.12	0.67	0.16	1.80
8	1.27	0.90	0.00	-0.03	0.04	0.76	434.65	0.69	0.18	2.01
9	1.44	0.85	-0.01	-0.05	0.06	0.76	418.20	0.70	0.20	2.19
10 (High)	1.89	0.77	-0.01	-0.08	0.09	0.76	339.86	0.80	0.20	5.58

Panel B: Correlations

	AB_NR	AB_PR	RET	RETOC_M	RETCO_M	NH	ME	BM	MOM	ILLIQ
AB_NR	1									
AB_PR	-0.265	1								
RET	-0.076	0.125	1							
RETOC_M	-0.179	0.212	0.338	1						
RETCO_M	0.310	-0.291	0.324	-0.235	1					
NH	0.002	0.048	0.267	0.114	-0.030	1				
ME	-0.002	-0.003	0.002	-0.005	0.003	0.099	1			
BM	0.007	0.016	0.031	0.026	-0.021	0.010	-0.063	1		
MOM	0.061	-0.016	-0.015	-0.018	0.002	0.350	0.011	0.070	1	
ILLIQ	-0.013	-0.010	0.003	0.023	-0.032	-0.049	-0.013	0.053	-0.029	1

Preprint not peer reviewed

This preprint research paper has not been peer reviewed. Electronic copy available at: https://ssrn.com/abstract=4482812

Table 2. Portfolio Returns sorted by AB_NR

This table represents the raw and risk-adjusted returns for the AB_NR premium from Akbas et al. (2022). For each month t, individual stocks are ranked into deciles based on their AB_NR values calculated in month t–1. Each decile portfolio is held for month t with equal or value weights. The AB_NR premium is the difference in returns between the lowest and highest deciles. We regress the AB_NR premium on two sets of factor models to obtain risk-adjusted returns. The returns are reported in percentages. The t-statistics calculated using Newey and West's (1987) robust standard errors with lag 12 are shown in parentheses. The sample period is from May 1993 to December 2020.

	Equal-weighted returns			Value-weighted returns		
AB_NR decile	Raw return	4-factor alpha	5-factor alpha	Raw return	4-factor alpha	5-factor alpha
1 (Low)	0.55	-0.04	-0.17	0.69	0.14	-0.07
2	0.57	-0.12	-0.22	0.66	-0.01	-0.05
3	0.73	0.01	-0.06	0.64	-0.08	-0.12
4	0.72	-0.01	-0.06	0.84	0.18	0.07
5	0.78	0.02	-0.01	0.71	-0.05	-0.07
6	0.76	-0.01	-0.03	0.72	-0.04	-0.04
7	0.93	0.15	0.10	0.84	0.09	0.13
8	1.04	0.27	0.21	0.78	0.02	0.05
9	1.15	0.37	0.33	0.83	0.09	0.13
10 (High)	1.37	0.66	0.57	0.95	0.22	0.07
10 – 1	0.83	0.70	0.74	0.27	0.09	0.14
	(5.37)	(5.02)	(4.78)	(1.27)	(0.46)	(0.67)

Preprint not peer reviewed

This preprint research paper has not been peer reviewed. Electronic copy available at: https://ssrn.com/abstract=4482812

Table 3. Double-sort Analysis

This table presents the AB_NR premia conditioned on nearness to the 52-week high price (NH). We split the sample of all stocks into quintile portfolios based on whether the NH. Next, we independently divide individual stocks into quintiles according to their AB_NR values and calculate equally-weighted returns for every 25 portfolios. We then compute the AB_NR premia separately for each NH quintile by taking the differences in returns between the lowest and highest AB_NR quintiles. The returns are reported in percentages. Numbers in parentheses are the t-statistics calculated using Newey and West's (1987) robust standard errors with lag 12. The sample period is from May 1993 to December 2020.

Equal-Weighted Returns

		AB_NR quintile							
		1 (Low)	2	3	4	5 (High)	5 – 1	4-factor alpha	5-factor alpha
NH quintile	1 (Far)	0.07	0.28	0.27	0.83	1.14	1.08	0.93	1.06
							(5.50)	(4.63)	(4.96)
	2	0.31	0.57	0.63	0.88	1.26	0.95	0.87	0.89
							(6.23)	(5.96)	(5.87)
	3	0.68	0.76	1	1.01	1.37	0.69	0.63	0.65
							(5.61)	(5.42)	(4.74)
	4	0.78	1.02	1.03	1.16	1.31	0.54	0.45	0.52
							(4.65)	(4.37)	(4.24)
	5 (Near)	0.82	1.01	1.05	1.06	1.15	0.33	0.28	0.33
							(2.83)	(2.40)	(2.42)
	5 – 1						-0.75	-0.66	-0.74
							(-4.23)	(-3.26)	(-3.29)

Preprint not peer reviewed

This preprint research paper has not been peer reviewed. Electronic copy available at: https://ssrn.com/abstract=4482812

Table 4. Fama and MacBeth (1973) Regression

This table presents the times-series average results of Fama-MacBeth (1973) firm-level cross-sectional regressions for estimating the relationships between stock returns and various factors. For each model, we run the following regression equation:

$$RET_{i,t+1} = \lambda_{0,t} + \lambda_{1,t} \cdot AB_NR_{i,t} + \lambda_{2,t} \cdot NH_{i,t} + \lambda_{3,t} \cdot AB_NR_{i,t} \times NH_{i,t} + \psi_t \cdot Z_{i,t}$$

where the dependent variable is the stock's return ($RET_{i,t+1}$), and the independent variables include the stock's AB_NR ($AB_NR_{i,t}$), near their 52-week high ($NH_{i,t}$), and the interaction term between AB_NR and the near their 52-week high variable ($AB_NR_{i,t} \times NH_{i,t}$). Additionally, $Z_{i,t}$ is the vector of several other control variables, including market beta, past returns, market capitalization, book-to-market ratio, illiquidity, turnover, and idiosyncratic volatility. The detailed definition of control variables is in Appendix A. The t-statistics calculated using Newey and West's (1987) robust standard errors with lag 12 are reported in parentheses. Sample period is from May 1993 to December 2020.

Variable	Model 1	Model 2	Model 3	Model 4	Model 5	Model 6
AB_NR	0.205	0.139	0.183	0.150	0.173	0.142
	(6.35)	(6.26)	(6.84)	(6.30)	(6.50)	(5.92)
AB_PR	-0.032	-0.018	-0.006	-0.017	0.003	-0.012
	(-0.90)	(-0.65)	(-0.22)	(-0.71)	(0.12)	(-0.53)
RETOC_M	-0.234	-0.319	-0.455	-0.365	-0.409	-0.263
	(-2.80)	(-4.26)	(-4.92)	(-5.31)	(-4.57)	(-4.13)
RETCO_M	-0.205	-0.146	-0.284	-0.181	-0.172	-0.021
	(-2.58)	(-3.58)	(-3.42)	(-3.64)	(-1.96)	(-0.37)
AB_NR × NH			-0.088	-0.083	-0.120	-0.113
			(-4.92)	(-4.36)	(-5.09)	(-4.88)
AB_PR × NH			0.017	0.023	0.028	0.031
			(0.96)	(1.31)	(1.22)	(1.46)
RETOC_M					0.150	0.193
× NH					(2.53)	(3.69)
RETCO_M					0.234	0.266
× NH					(5.53)	(7.27)
NH			0.427	0.121	0.405	0.048
			(3.01)	(1.06)	(2.87)	(0.40)
Controls	N	Y	N	Y	N	Y

Preprint not peer reviewed

This preprint research paper has not been peer reviewed. Electronic copy available at: https://ssrn.com/abstract=4482812

Table 5. Investor Sentiment and Double-sort Analysis

This table presents the AB_NR return spreads conditioned on nearness to the 52-week high price (NH) for high and low sentiment periods. The high (low) sentiment periods are when the investor sentiment index from Baker and Wurgler (2006) exceeds (falls below) the median value. We perform a double-sort analysis for high and low sentiment periods, respectively. We split the sample of all stocks into quintile portfolios based on whether the NH. Next, we independently divide individual stocks into quintiles according to their AB_NR values and calculate equally-weighted returns for every 25 portfolios. We then compute the AB_NR premia separately for each NH quintile by taking the differences in returns between the lowest and highest AB_NR quintiles. The returns are reported in percentages. Numbers in parentheses are the t-statistics calculated using Newey and West's (1987) robust standard errors with lag 12. The sample period is from May 1993 to December 2020.

	High sentiment		Low sentiment	
	AB_NR return spreads		AB_NR return spreads	
NH quintile	4-factor alpha	5-factor alpha	4-factor alpha	5-factor alpha
1 (Far)	1.29	1.48	0.68	0.71
	(4.20)	(5.34)	(2.79)	(2.79)
2	1.10	1.19	0.63	0.62
	(5.00)	(5.26)	(3.34)	(3.24)
3	0.79	0.86	0.45	0.45
	(4.88)	(5.06)	(2.83)	(2.59)
4	0.56	0.60	0.36	0.42
	(3.82)	(3.71)	(2.74)	(2.79)
5 (Near)	0.35	0.37	0.20	0.27
	(2.03)	(2.04)	(1.54)	(1.87)
5 – 1	-0.94	-1.11	-0.48	-0.44
	(-3.14)	(-3.94)	(-1.72)	(-1.59)

Preprint not peer reviewed

This preprint research paper has not been peer reviewed. Electronic copy available at: https://ssrn.com/abstract=4482812